2. Explain the difference between method declaration and method body.
Method declaration specifies the method's name, return type, access modifier, and parameters
e.g. public static int add(int int1, int int2)

Method body contains the code inside the curly braces that performs the task
e.g. {
    return int1 + int2;
}

3. What type of keyword is used to change the access level of a method.
An access modifier, such as public or private.

4. What is another word used for describing the access level of a method?
Visibility.

- Public method allows method to be accessed from other classes
- Private method restricts access to within the same class

5. Explain the scope of each of the variables in the code below:

```
public class ScopeExample {

        public static void method1()  {
                int var3;
                for (int var4 = 0; var4 < 2; var4++) {
                        var3 += 1;
                }
        }
}

public static void main(String[] args) {
        int var1;
        for (int var2 = 0; var2 < 5; var2++) {
                method1();
        }
}
```

var1: a local variable that can only be accessed in main()
var2: a local variable defined in the for loop in main(), it can only be accessed within that loop.
var3: a local variable that can only be accessed in method1(), also it is not initialized in this code therefore an error would occur
var4: a local variable defined in the for loop in method1(), it can only be accessed within that loop.

6. Write a method declaration for each of the following descriptions:

a) A class method named getVowels that can be called by any other method, requires a String parameter, and returns an integer value.
public static int getVowels(String word)
b) A class method named extractDigit that can be called by any other method, requires an integer parameter, and returns an integer value.
public static int extractDigit(int number)
c) A class method named insertString that can be called by any other method, requires a String parameter and an integer parameter, and returns a String parameter.
public static String insertString(String word, int number)

7.

a) How does the compiler distinguish one method from another?
By their method name, numbers, types, and order of its parameters.
b) Can two methods in the same class have the same name? Explain.
Yes, this is called method overloading. They must have different parameter lists, e.g.
public static int add(int a, int b)
Or
public static int add(int a, int b, int c)

8.

a) What is the return statement used for?
It ends a method and sends the value back to the calling statement if the method returns a value
b) How many values can a return statement send back to a calling statement?
One value. To return multiple values requires an array or an object.
c) How is the declaration of a method returning a value different from the declaration of a method that does not return a value?
A method returning a value specifies the return type, such as int, or String. A method that does not return a value uses void. However, a void method can still use return statements without returning a value.

9. Find and explain the error in the code below:

```
public class MethodCallExample {

        public static int doSomething() {
                return(5);
        }
}

public static void main(String[] argos) {
        int num;
        doSomething();
        num = doSomething();
```

}

There is nothing catching the return value for doSomething() in the highlighted code.

11. Determine if each of the following are true or false. If false, explain why.
    a) Breaking a task down into methods is called procedural abstraction. true
    b) A method call consists of the method declaration in an assignment statement. False, a method call invokes a method
    c) A void method must return a value. False, void methods do not have to return values
    d) An access modifier declares the return type of a method False, an access modifier determines visibility, not type
    e) The keyword static declares a method is a class method true
    f) Method parameters are enclosed by braces {}. False, method parameters are enclosed by parentheses (), not braces {}
    g) Local variables can be used by any method in a class. False, local variables can only be used within their declared scope
    h) The value of an argument passed to a method can be changed in an assignment statement in the method. False, reassigning a parameter does not change the caller's original value.
    i) Method overloading means that an application contains more than 10 methods. False, method overloading means using the same method name with different parameters.
    j) The return statement is used to send a value back to the calling statement. true
    k) The precondition of a method states the data types of the method's parameters. False, a precondition describes what must be true before the method runs
    l) The postcondition of a method describes the way the method accomplishes its task. False, a postcondition describes what is true after the method has finished, not how it accomplishes its task.